A Certified Robustness Threshold for Constructible Cubic Angle Approximation

Deterministic Error Bounds, Statistical Tolerance Certification, and Measurement Uncertainty

C. Wayne Baker
Independent researcher, Goulds, Newfoundland and Labrador, Canada

5 November 2026 — manuscript version 2.3

Abstract

Exact angle trisection by straightedge and compass is impossible in general, so practical constructions approximate $\theta/3$. For the publicly documented Baker seed $D_4(\theta) = 2\arcsin\left(\frac{4}{3}\sin\frac{\theta}{8}\right)$, which overshoots $\theta/3$ on $0 < \theta \le 120^\circ$ with leading error $\frac{7}{10368}\theta^3$, we ask when that overshoot remains *guaranteed* once the construction is carried out with imperfect instruments. We prove an exact necessary-and-sufficient chord-error tolerance $E_4(\theta) = 2\sin\frac{\theta}{8} - \frac{3}{2}\sin\frac{\theta}{6}$, extend it to simultaneous chord, scaling, calibration and arc-transfer errors, and show by convexity that the certified domain is always a single interval $(\theta_{\min}, 120^\circ]$. Fixed absolute errors give $\theta_{\min} \sim 24\,7^{-1/3}\varepsilon_c^{1/3}$; relative calibration error gives $\theta_{\min} \sim \sqrt{3456r/7}$. We then separate this deterministic certificate from a statistical one: a random-effects calibration model, a readout-transfer theorem that converts tolerance bounds on measured setup means into bounds on future construction errors, a Lipschitz envelope for certification over a continuous angular interval, and Bonferroni confidence accounting. Reproducible Monte Carlo benchmarks (four frozen versions) show that Gaussian tail certificates become *more* confidently wrong under heavy-tailed errors as calibration data grow, that naive transfer from setup means fails on an explicit counterexample, and that the corrected distribution-free procedure is valid at a quantified calibration cost. No claim of exact trisection is made, and the statistical methods used are standard; the contribution is their exact, construction-specific error budget.

Contents

1 Introduction **2**

2 The public D_4 construction **3**

3 Exact deterministic threshold for chord error **3**

4 Multistage measurement errors **3**
4.1 Design rule: radius and instrument gates 4

5 Repeated constructions and random effects **5**

6 Tolerance bounds and readout transfer **6**

7 Continuous-angle certification **6**

8 Reproducible experiments **7**

9 Calibration cost versus population coverage **8**

10 Discussion and limitations **8**

A Interval enclosures **9**

B Reproducibility **9**

C Source and output hashes **10**

1 Introduction

Wantzel showed in 1837 that a general angle cannot be trisected with straightedge and compass [2]. That negative answer leaves open a practical question that every surveyor will recognise: an approximate construction is useful only if one can say, *before* knowing the exact answer, how far it can be trusted.

This paper studies one public construction, the overshooting seed D_4 of [1], and asks:

> For which input angles can we *prove* that the physically executed construction still overshoots $\theta/3$, given stated bounds on construction and measurement error?

The answer has two forms that must not be confused. A *deterministic certificate* (Sections 3–4) holds for every error within hard bounds. A *statistical certificate* (Sections 5–7) holds for a specified fraction of future constructions, with a specified confidence, under explicit calibration assumptions. Neither is a proof of exact trisection.

Related work and scope of novelty. Approximate angle trisection has been investigated previously, including rigorous error analysis by Dacar (2011) [13] and earlier work published in *Survey Review* by Porter (1991) [14]. Error propagation under uncertain geometric data has also been studied by Wallner, Krasauskas and Pottmann (2000) [15], using a geometric framework involving exact tolerance zones and Laguerre geometry, and by Desaulniers and Stewart (1993) [16], as broader background in robust geometric computation. The present contribution is narrower: for the publicly released Baker D_4 construction [1], we derive an exact admissible chord-error threshold, extend it to a multistage measurement-error budget, and formulate conditional statistical certification over a continuous interval of input angles. No priority claim is made for approximate trisection, geometric error propagation, or the classical statistical methods employed.

Contributions. (i) An exact iff tolerance for the input chord (Theorem 2) and for combined multistage errors (Theorem 3), with a convexity proof that the certified set is a single interval (Theorem 4). (ii) Two threshold scaling laws and an explicit radius/instrument design rule (Section 4.1). (iii) A readout-transfer theorem with a counterexample showing why it is needed (Section 6). (iv) A Lipschitz envelope and an exact, grid-free threshold algorithm (Section 7). (v) Frozen, hashed benchmarks quantifying when each certificate is valid and what it costs (Sections 8–9). The scaling laws are the generic consequence of a cubic bias competing with an error floor; the statistical tools (random-effects ANOVA, tolerance intervals, Bonferroni, Clopper–Pearson) are classical. What is specific to this paper is the exact error budget for this Euclidean construction.

2 The public D_4 construction

On a unit circle let $c(\theta) = 2\sin(\theta/8)$ be the chord subtending $\theta/4$. The public seed [1, Sec. 3] is

$$D_4(\theta) = 2\arcsin\Big(\tfrac{4}{3}\sin\tfrac{\theta}{8}\Big) = 2\arcsin\Big(\tfrac{2}{3}\,c(\theta)\Big), \tag{1}$$

obtained by scaling a constructed chord by the rational factor $2/3$ and transferring the resulting half-chord back to an angle. Write $e_4(\theta) = D_4(\theta) - \theta/3$. Then $e_4(\theta) = \frac{7}{10368}\theta^3 + O(\theta^5)$ [1].

Lemma 1 (Noiseless overshoot and convexity)**.** *On $0 < \theta \le 120°$: (a) $e_4(\theta) > 0$; (b) $e_4''(\theta) > 0$.*

Proof. (a) Since $\theta/6 \le 20°$ and arcsin is increasing, $e_4 > 0$ iff $\frac{2}{3}c > \sin\frac{\theta}{6}$, i.e. iff $E_4(\theta) := 2\sin\frac{\theta}{8} - \frac{3}{2}\sin\frac{\theta}{6} > 0$. $E_4(0) = 0$ and $E_4'(\theta) = \frac{1}{4}(\cos\frac{\theta}{8} - \cos\frac{\theta}{6}) > 0$ because $0 < \frac{\theta}{8} < \frac{\theta}{6} < \pi$. (b) With $u = \theta/8$, $k = 4/3$ and $f(u) = \arcsin(k\sin u)$, $f''(u) = k(k^2-1)\sin u\,(1-k^2\sin^2 u)^{-3/2} > 0$ for $k > 1$, $0 < u \le 15°$; and $e_4''(\theta) = \frac{2}{64}f''(u)$. □

3 Exact deterministic threshold for chord error

Suppose the input chord is realised as $c + \delta_c$ with $|\delta_c| \le \varepsilon_c$, all else exact.

Theorem 2 (Exact chord tolerance)**.** *The perturbed construction overshoots $\theta/3$ for every admissible δ_c if and only if $\varepsilon_c < E_4(\theta)$. E_4 is strictly increasing on $(0, 120°]$ with $E_4(\theta) = \frac{7}{13824}\theta^3 + O(\theta^5)$, so for $0 < \varepsilon_c < E_4(120°)$ there is a unique $\theta_{\min}(\varepsilon_c)$ and the guarantee holds exactly on $(\theta_{\min}, 120°]$, with*

$$\theta_{\min}(\varepsilon_c) \sim \frac{24}{7^{1/3}}\,\varepsilon_c^{1/3} \approx 12.5462\,\varepsilon_c^{1/3} \qquad (\varepsilon_c \to 0).$$

For $\varepsilon_c \ge E_4(120°) \in [0.0046078752165,\ 0.0046078752166]$ no angle in the domain is certified.

Proof. The worst case is $\delta_c = -\varepsilon_c$ because arcsin is increasing; arguments stay on the principal branch since $\frac{2}{3}(c + \varepsilon_c) < 1$ for all relevant ε_c. The inequality $2\arcsin(\frac{2}{3}(c - \varepsilon_c)) > \theta/3$ is equivalent to $\varepsilon_c < E_4(\theta)$ as in Lemma 1(a); monotonicity is Lemma 1(a); the constant follows from $\frac{7}{13824} = 7/24^3$. □

The chord-to-angle sensitivity is $\partial D_4/\partial c = \frac{4}{3}\,(1 - \frac{16}{9}\sin^2\frac{\theta}{8})^{-1/2}$, equal to $4/3$ at 0 and ≈ 1.4206026 at $120°$.

ε_c	certified enclosure of $\theta_{\min}$ (deg)	cubic estimate (deg)
10^{-8}	[1.548702823, 1.548702826]	1.548702
10^{-6}	[7.188519788, 7.188519791]	7.188438
10^{-4}	[33.373964194, 33.373964197]	33.365773

Table 1: Robust-overshoot thresholds for input-chord error (unit radius). Enclosures are outward-rounded interval evaluations (Appendix A); the cubic column is the leading-order estimate.

4 Multistage measurement errors

Model the executed construction as

$$\widetilde{D}_4 = 2\arcsin\Big[\tfrac{2}{3}(1+\eta)(c+\delta_c) + \delta_s\Big] + \delta_a,$$

with $|\delta_c| \le \varepsilon_c$ (chord), $|\delta_s| \le \varepsilon_s$ (scaled length), $|\eta| \le r < 1$ (relative calibration of the factor 2/3) and $|\delta_a| \le \varepsilon_a$ (final arc transfer, radians).

Theorem 3 (Combined error budget)**.** *Assume the four errors range independently over their boxes (so the all-adverse corner is attainable), all arguments remain on the principal branch, and $c(\theta) \ge \varepsilon_c$, so every perturbed chord $c + \delta_c$ is nonnegative. The construction is a guaranteed strict overshoot iff $F(\theta) > 0$, where*

$$F(\theta) = (1-r)\Big(2\sin\tfrac{\theta}{8} - \varepsilon_c\Big) - \tfrac{3}{2}\varepsilon_s - \tfrac{3}{2}\sin\Big(\tfrac{\theta}{6} + \tfrac{\varepsilon_a}{2}\Big).$$

If the errors are coupled, $F > 0$ remains sufficient but need not be necessary. Where $c(\theta) < \varepsilon_c$ the perturbed chord can vanish, the constructed angle can be near $0 < \theta/3$, and no overshoot is guaranteed; this regime lies below any certified interval.

Proof. Since $c+\delta_c \ge 0$, the argument $\frac{2}{3}(1+\eta)(c+\delta_c)+\delta_s$ is nondecreasing in each of η, δ_c, δ_s, so the worst corner is $\eta = -r$, $\delta_c = -\varepsilon_c$, $\delta_s = -\varepsilon_s$, $\delta_a = -\varepsilon_a$; $2\arcsin(\cdot) - \varepsilon_a > \theta/3$ iff $\arcsin(\cdot) > \theta/6 + \varepsilon_a/2$, and $\theta/6 + \varepsilon_a/2 < \pi/2$ allows taking sines. □

Theorem 4 (Single threshold)**.** *On $(0, 120^\circ]$, $F''(\theta) = \frac{1}{24}\sin(\frac{\theta}{6} + \frac{\varepsilon_a}{2}) - \frac{1-r}{32}\sin\frac{\theta}{8} > 0$, and $F(0) = -(1-r)\varepsilon_c - \frac{3}{2}\varepsilon_s - \frac{3}{2}\sin\frac{\varepsilon_a}{2} \le 0$. If $\varepsilon_c = \varepsilon_s = \varepsilon_a = 0 < r$, then $F(0) = 0$ and $F'(0) = -r/4 < 0$. In either case, whenever some error is nonzero, $F < 0$ on some interval $(0, \tau)$. Hence either $F(120^\circ) \le 0$ and nothing is certified, or there is exactly one root $\theta_{\min}$ and the certified set is $(\theta_{\min}, 120^\circ]$.*

Proof. $\sin(\theta/6 + \varepsilon_a/2) \ge \sin(\theta/6) > \sin(\theta/8)$ and $1/24 > (1-r)/32$. A strictly convex function that is negative on $(0, \tau)$ has at most one zero in $(0, 120^\circ]$: if it had two, $\theta_1 < \theta_2$, convexity on $[\tau/2, \theta_2]$ with $F(\tau/2) < 0 = F(\theta_1)$ would force $F(\theta_2) > 0$. With all errors zero, $F = E_4 > 0$ by Lemma 1. □

Near 0 the margin is $\frac{7}{13824}\theta^3 - \frac{r}{4}\theta - \varepsilon_c - \frac{3}{2}\varepsilon_s - \frac{3}{4}\varepsilon_a + \dots$, so pure calibration error gives $\theta_{\min} \sim \sqrt{3456r/7}$, and the guarantee is lost everywhere once

$$r \;\ge\; E_4(120^\circ)/c(120^\circ) \;\in\; [0.008901731,\, 0.008901732].$$

4.1 Design rule: radius and instrument gates

Errors fall into three classes: fixed absolute lengths (scale as $1/R$ on a drawing of radius R), fixed relative calibration (independent of R), and fixed angular errors (independent of R unless they arise from endpoint displacement). Enlarging the drawing therefore helps only the first class. For the illustrative hand budget 0.30 mm chord, 0.15 mm scale, 0.1% calibration and 0.10° arc, $F(120^\circ) < 0$ at $R = 100$ mm, and the critical radius is enclosed in [183.4373024, 183.4373027] mm. As $R \to \infty$ the margin is positive only if the arc gate

$$\varepsilon_a < 2\arcsin\Big(\tfrac{4}{3}(1-r)\sin 15^\circ\Big) - 40^\circ$$

holds; for $r = 10^{-3}$ this is $\varepsilon_a < 0.3327^\circ$ (high-precision evaluation, not interval-certified). No drawing size compensates for arc transfer worse than this.

Model	ε_c	ε_s	ε_a	r	enclosure of $\theta_{\min}$ (deg)
Fine numerical/CAD	10^{-8}	10^{-9}	10^{-8} rad	10^{-7}	[1.9463246, 1.9463249]
Controlled drafting (R = 100 mm)	10^{-4}	5×10^{-5}	0.01°	10^{-4}	[49.5050435, 49.5050438]
Coarse drafting (R = 100 mm)	5×10^{-4}	2×10^{-4}	0.05°	10^{-3}	[87.8108575, 87.8108578]

Table 2: Certified thresholds for illustrative combined budgets (hypothetical tolerances, not instrument specifications).

5 Repeated constructions and random effects

Deterministic certificates need hard bounds; in practice those come from calibration. Construct the angle independently m times and read each construction k times:

$$Y_{ij} = D_4(\theta) + b + U_i + V_{ij}, \qquad \mathrm{Var}(U_i) = \sigma_C^2,\ \mathrm{Var}(V_{ij}) = \sigma_R^2,$$

with b a persistent bias. Then $\mathrm{Var}(\overline{Y}) = \sigma_C^2/m + \sigma_R^2/(mk)$: readings average out readout noise only; setup-to-setup variation needs new constructions. In a balanced design the variance of $\overline{Y}$ is estimated by $\widehat{\mathrm{Var}}(\overline{Y}) = MS_{\text{between}}/(mk)$, and the estimated standard error is its square root. The between-setup mean square has $m - 1$ degrees of freedom [6]; Satterthwaite's approximation [7] reduces to this case. Treating all mk readings as independent is pseudo-replication. The variance-component decomposition and its implications for allocating independent constructions and repeated readings follow from the present model and standard variance-components theory [6]; geodetic network optimization offers related practical context [12]. Table 4 quantifies the trade-off for this construction.

Remark 5 (Role of the experiment)**.** If the constructed angle can be measured to arcseconds, so can θ. The repeated-construction experiment is therefore a *validation* study that characterises an apparatus; the deterministic theorem is then applied to constructions performed without measurement.

Remark 6 (An irony)**.** The arithmetic mean of 2^j constructible angles is constructible by addition and repeated bisection, but averaging three arbitrary constructions would require trisecting their sum. Experimental averages here are numerical.

Scenario	m	exact t	z	naive iid z	Gaussian tolerance
bias	4	94.97%	90.13%	76.67%	95.36%
bias	32	95.03%	94.50%	80.12%	96.23%
control	4	94.97%	90.07%	76.56%	95.26%
control	32	95.06%	94.56%	80.47%	96.22%
heavy	4	95.59%	90.51%	76.43%	73.82%
heavy	32	95.02%	94.51%	80.10%	9.87%
stage	4	94.96%	89.97%	76.40%	95.30%
stage	32	95.03%	94.51%	80.37%	96.08%

Table 3: v1: empirical coverage of nominal one-sided 95% procedures, averaged over angles and k. Gaussian tolerance = 99.9%/95% normal tolerance bound.

constructions m	readings k	median $\theta_{\min}$ (deg)
16	2	39.55
8	4	41.99
4	8	47.96

Table 4: v1, Gaussian control, $\theta = 60°$, 32 observations: independent constructions beat repeated readings.

6 Tolerance bounds and readout transfer

The certificate needs a lower bound L_W with $P(W \le L_W) \le \alpha$ for a *future* construction error $W = b + U$, whereas calibration observes setup means $X = W + \overline{V}$.

Proposition 7 (Naive transfer can fail)**.** *Let $W = -A$ with probability α and 0 otherwise, and let $\overline{V}$ be symmetric with $P(\overline{V} \le 0) = \frac{1}{2}$ and support in $[-d, d]$, $0 < d < A$. Then the lower α-quantile of W is $-A$, but $P(X \le -A) = \alpha/2 < \alpha$, so the lower α-quantile of X exceeds $-A$. A tolerance bound valid for X can lie above the true α-quantile of W.*

Theorem 8 (Readout transfer)**.** *If $P(X \le L_X) \le \alpha - \beta$ and $P(\overline{V} > \delta) \le \beta$, then $L_W = L_X - \delta$ satisfies $P(W \le L_W) \le \alpha$.*

Proof. $\{X - \overline{V} \le L_X - \delta\} \subseteq \{X \le L_X\} \cup \{\overline{V} > \delta\}$. □

The allowance δ must not be estimated from within-setup residuals $V_{ij} - \overline{V}_i$, which are dependent and are not draws of $\overline{V}$. In v2.2 it is the maximum of an *independent* reference experiment in which blocks of k readings of a known reference angle give i.i.d. draws of $\overline{V}$; this requires the readout distribution to be the same in reference and calibration runs.

Distribution-free bounds use order statistics [3]: the r-th smallest of m i.i.d. observations lies below the p-quantile with probability $P(\mathrm{Bin}(m, p) \ge r)$; for $r = 1$ the confidence is $1 - (1 - p)^m$. Parametric normal bounds use noncentral-t factors [4, 5]. With five calibration angles and one readout bound, Bonferroni allocation [8] at $1 - 0.05/6$ per event gives at least 95% simultaneous confidence without independence assumptions.

7 Continuous-angle certification

Bounds at node angles $\theta_1 < \cdots < \theta_5$ do not automatically cover angles in between.

Theorem 9 (Lipschitz envelope)**.** *Suppose every realisation satisfies $|W(\theta) - W(\phi)| \le \Lambda|\theta - \phi|$. Then the lower α-quantile $q(\theta)$ is Λ-Lipschitz, and on the event that all node bounds $L_j \le q(\theta_j)$ hold,*

$$L(\theta) = \max_j \left(L_j - \Lambda|\theta - \theta_j|\right) \le q(\theta) \qquad \text{for all } \theta \in [\theta_1, \theta_5].$$

Proof. Pathwise Lipschitz continuity shifts the distribution function by at most $\Lambda|\theta - \phi|$, hence the quantile; then $q(\theta) \ge q(\theta_j) - \Lambda|\theta - \theta_j| \ge L_j - \Lambda|\theta - \theta_j|$ for every j. □

For purely angular error models W does not depend on θ and $\Lambda = 0$ (a modelling hypothesis). For the physical stage model, Gaussian errors admit no pathwise Λ; truncating them at 4σ gives $|\partial W/\partial\theta| \le 2x'_{\max}[\eta_{\max} s_{\max} + s'_{\max}\,\Delta a_{\max}] = 5.17 \times 10^{-5}$, with $x'_{\max} = 1/6$, $s(z) = (1 - z^2)^{-1/2}$ and $\Delta a_{\max}$ the worst-case shift of the arcsine argument; a finite-difference check gives a maximum of 3.90×10^{-5}.

Scenario	method	m	simultaneous node coverage	wrong certificates	median $\theta_{\min}$
control	P	4	95.11%	1/10000	63.00°
control	P	32	97.43%	0/10000	40.50°
control	NP	4603	99.80%	0/2000	37.75°
control	NP	8000	99.85%	0/2000	37.25°
bias	P	4	95.48%	0/10000	63.75°
bias	P	32	97.32%	0/10000	42.00°
bias	NP	4603	99.75%	0/2000	39.25°
bias	NP	8000	100.00%	0/2000	38.75°
heavy	P	4	72.41%	25/10000	61.25°
heavy	P	32	0.10%	5945/10000	40.25°
heavy	NP	4603	95.85%	1/2000	51.25°
heavy	NP	8000	98.90%	0/2000	48.75°
stage	P	4	95.46%	1/10000	64.75°
stage	P	32	97.08%	0/10000	41.25°
stage	NP	4603	98.75%	0/2000	38.25°
stage	NP	8000	99.85%	0/2000	37.75°

Table 5: v2.1 (Bonferroni, 0.99 per node, readout transfer not yet applied). Under heavy tails the Gaussian procedure becomes more confidently wrong as m grows.

Exact threshold algorithm. L is piecewise linear with breakpoints at the nodes and pairwise crossings. On each piece $M = e_4 + L$ is strictly convex (Lemma 1b) with slope $e_4' + s$, $s \in \{-\Lambda, 0, \Lambda\}$, so its minimum is at the left end ($s \geq 0$) or at θ_c solving $e_4'(\theta_c) = \Lambda$. Scanning pieces from 120° downward and solving one root gives the final certified interval exactly; no grid is used. The returned root is a binary64 value; Appendix A gives interval enclosures for the deterministic thresholds.

8 Reproducible experiments

All benchmarks use only the public D_4, binary64 arithmetic and fixed seeds, and are frozen with SHA-256 hashes (Appendix C). Illustrative values: $\sigma_C = 10''$, $\sigma_R = 4''$, $k = 2$ (v2.x). Scenarios: Gaussian control; bias $b = -5''$; Student-t_4 errors scaled to the same SD; physical stage errors propagated through the exact nonlinear map (truncated at 4σ in v2.2); and, in v2.2, the atom counterexample of Proposition 7 ($A = 60''$, uniform readout half-width $20''$). Validity is scored against the true distribution of W (analytic, or 2×10^6-sample simulation for the stage model) on a 0.05° grid; this scoring grid is not part of the certificate.

- **v1**: correct t inference holds 95%; z undercovers at small m; pseudo-replication gives 77–80%; Gaussian tolerance bounds fail under heavy tails, increasingly with m (Table 3).
- **v2/v2.1**: five calibration angles, simultaneous accounting, order-statistic bounds; v2.1 adopts Bonferroni (Table 5).
- **v2.2**: readout transfer, Lipschitz envelope, exact threshold, atom regression test (Table 6). The regression test confirms that naive transfer fails (58.05% simultaneous coverage) and corrected transfer passes.

Two observations deserve emphasis. First, using all nodes in the envelope makes it sharp enough that the Gaussian method in the Gaussian control shows 2.5–4.2% wrong certificates: within its 5% budget, but no longer the near-zero rate of the more conservative v2.1 envelope. Second, the naive order-statistic method passes the smooth scenarios; only the counterexample exposes it, which is why it is kept as a permanent regression test.

Scenario	method	m	simult. cov.	wrong / runs	95% upper	$\theta_{\min}$	truth
control	Gaussian	8	95.74%	419/10000	4.53%	41.21°	34.65°
control	Gaussian	32	97.32%	254/10000	2.81%	37.40°	34.65°
control	NP naive	4603	99.65%	6/2000	0.59%	35.96°	34.65°
control	NP transfer	5318	100.00%	0/2000	0.15%	39.78°	34.65°
bias	Gaussian	8	95.82%	417/10000	4.51%	42.56°	36.45°
bias	Gaussian	32	97.32%	263/10000	2.91%	38.94°	36.45°
bias	NP naive	4603	99.65%	7/2000	0.66%	37.61°	36.45°
bias	NP transfer	5318	100.00%	0/2000	0.15%	41.15°	36.45°
heavy	Gaussian	8	28.63%	7090/10000	71.65%	39.01°	40.85°
heavy	Gaussian	32	0.09%	9990/10000	99.95%	36.06°	40.85°
heavy	NP naive	4603	95.80%	79/2000	4.74%	43.97°	40.85°
heavy	NP transfer	5318	100.00%	0/2000	0.15%	51.74°	40.85°
stage4	Gaussian	8	95.89%	142/10000	1.63%	43.62°	35.15°
stage4	Gaussian	32	97.32%	25/10000	0.35%	39.50°	35.15°
stage4	NP naive	4603	99.20%	0/2000	0.15%	37.67°	35.15°
stage4	NP transfer	5318	100.00%	0/2000	0.15%	41.28°	35.15°
atom	Gaussian	8	5.14%	9477/10000	95.13%	39.04°	43.20°
atom	Gaussian	32	0.00%	10000/10000	100.00%	35.13°	43.20°
atom	NP naive	4603	58.05%	821/2000	42.89%	43.41°	43.20°
atom	NP transfer	5318	98.25%	35/2000	2.31%	47.96°	43.20°

Table 6: v2.2 results. "95% upper" is the Clopper–Pearson bound [9] on the wrong-certificate rate; "truth" is the true start of the certifiable interval. Every run issued a certificate. Median readout allowances δ: 11.8″ (Gaussian readout), 36.4″ (t_4), 19.9″ (atom).

9 Calibration cost versus population coverage

The distribution-free certificate is expensive, and the cost is driven by the coverage target $1-\alpha$ (Table 7, exact binomial minima).

Population coverage $1-\alpha$	$\beta=\alpha/10$	setups per node	total setups (5 nodes)	reference blocks	conf. per event
99.0%	0.001	530	2,650	4,786	$1-0.05/6$
99.5%	0.0005	1,062	5,310	9,573	$1-0.05/6$
99.9%	0.0001	5,318	26,590	47,873	$1-0.05/6$

Table 7: Minimum calibration sizes for the v2.2 procedure ($r=1$ order statistics, six Bonferroni events).

10 Discussion and limitations

- The deterministic theorems are exact for the stated additive box model; physical tolerances must be justified independently. A statistical tolerance does not become a worst-case guarantee.
- Statistical certificates are conditional on stationarity of the instrument between reference and calibration runs, on the Lipschitz hypothesis between nodes, and below 10° they are not

issued.

- Gaussian tail extrapolation should be used only with independent evidence for Gaussian tails; otherwise the order-statistic procedure and its calibration cost apply.
- Bias in the construction and bias in the measuring instrument cannot be separated without a calibrated reference.
- All σ values are illustrative; no physical instrument has yet been characterised.
- Monte Carlo reruns have not been performed independently of the author's environment; reported coverages carry Monte Carlo error (bounds in the CSV outputs).

A possible generalisation. Any constructible approximation with a strictly signed polynomial bias and a convex margin should admit an analogous unique robust threshold. We state this only as an open question; this paper establishes it for the public D_4 seed alone.

Acknowledgements. Derivations, code and audits were developed with the assistance of AI systems (Anthropic Claude and OpenAI ChatGPT) used as cross-checking collaborators; all results are reproducible from the released code, and responsibility rests with the author.

A Interval enclosures

Thresholds in Tables 1 and 2 and the critical radius were obtained by binary64 root finding, then rounding the root outward to the printed number of decimals. The *printed* decimal endpoints themselves were then certified: converted to radians as intervals, the functions $E_4 - \varepsilon_c$ or F were evaluated in outward-rounded interval arithmetic (`mpmath.iv`, 40 digits [11, 10]) and confirmed strictly negative at the lower endpoint and strictly positive at the upper. Uniqueness is supplied by Theorems 2 and 4, so each printed interval contains the unique root. The caps $E_4(120°)$ and $E_4(120°)/c(120°)$ are interval evaluations rounded outward. The arc gate of Section 4.1 was evaluated with ordinary 40-digit arithmetic, since `mpmath.iv` provides no interval arcsine; it is not interval-certified. Stochastic thresholds from the benchmarks are not interval-certified.

B Reproducibility

```
python3 d4_benchmark.py                 # v1   seed 20261104
python3 d4_benchmark_v2.py              # v2   seed 20261106
python3 d4_benchmark_v2_1.py            # v2.1 seed 20261125 (Bonferroni)
python3 d4_benchmark_v2_1.py --rule product
python3 d4_benchmark_v2_2.py            # v2.2 seed 20261126
cd paper && python3 make_paper_data.py  # regenerates every table in this paper
```

Environment recorded in the frozen benchmark logs (not independently re-created for this manuscript build): Python 3.13.18, NumPy 2.5.3, SciPy 1.18.1; mpmath 1.4.1 is used for the interval checks. The release archive contains the frozen benchmark scripts and outputs at its top level, with this paper in `paper/`; `make_paper_data.py` reads the frozen outputs from the parent directory.

C Source and output hashes

File	SHA-256
d4_benchmark.py	8a5b88b5425af3aa94e73979b26758964c2990a4ef9ae21e223ca96dddf07457
d4_benchmark_results.csv	69d3f1b8ef581aece832719eeb17aa26bc763103f4b973d4166381a3e2dbcca5
d4_benchmark_v2.py	58bbe6295f588a1be3b3197456fc87a9edde08d8cdfb66f378a26d5132ffc55a
v2_results.csv	4e9e3cada43cffd3820bcbde262d8cf24f0a8aee2d84be6a1228cbe672fd5bb4
d4_benchmark_v2_1.py	695f1cb1fff9ba8c5afad81e83a8503b7082be243eb16b2d8209e8118565c4fa
v21_bonferroni_results.csv	3bf41c163e01b003b93ad09e8245dc461082d4fbf6bd992ccd147df57cfa6aad
v21_product_results.csv	9cfc02d8d7bb6281d425feac94019a02762a9ae47161fd2b589fc1fc64019d68
d4_benchmark_v2_2.py	6f9f93ab78d6412871737e71217e9318ee3838c0d6351a7d3ad92b3da4e1f6ec
v22_results.csv	7534f09dfb9f01f2edee41869253b003b5ad23a17f8b5ee7b57d48076cc89028
v22_node_coverage.csv	ff6c4f67316c32fb071447db019711e1b3b23bfb3fb1c9bceccee0f3b3792d65
v22_log.json	225337fede7a783ff2f2b4aef91b6867042bfabb23e61a88107605cacfd3f5eb

References

[1] C. W. Baker, *A Constructible Cubic Error Cascade for Approximate Angle Trisection*, v1.0.1, https://github.com/CWBaker360/Constructible-Cubic-Trisection.

[2] P. L. Wantzel, Recherches sur les moyens de reconnaître si un problème de géométrie peut se résoudre avec la règle et le compas, *J. Math. Pures Appl.* 2 (1837) 366–372.

[3] S. S. Wilks, Determination of sample sizes for setting tolerance limits, *Ann. Math. Statist.* 12 (1941) 91–96.

[4] D. B. Owen, A survey of properties and applications of the noncentral t-distribution, *Technometrics* 10(3) (1968) 445–478.

[5] K. Krishnamoorthy and T. Mathew, *Statistical Tolerance Regions: Theory, Applications, and Computation*, Wiley, 2009, https://doi.org/10.1002/9780470473900.

[6] S. R. Searle, G. Casella and C. E. McCulloch, *Variance Components*, Wiley, 1992.

[7] F. E. Satterthwaite, An approximate distribution of estimates of variance components, *Biometrics Bulletin* 2 (1946) 110–114.

[8] O. J. Dunn, Multiple comparisons among means, *J. Amer. Statist. Assoc.* 56(293) (1961) 52–64.

[9] C. J. Clopper and E. S. Pearson, The use of confidence or fiducial limits illustrated in the case of the binomial, *Biometrika* 26(4) (1934) 404–413.

[10] R. E. Moore, R. B. Kearfott and M. J. Cloud, *Introduction to Interval Analysis*, SIAM, 2009.

[11] The mpmath development team, *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, version 1.4.1 (released 15 March 2026), https://pypi.org/project/mpmath/1.4.1/.

[12] E. W. Grafarend and F. Sansò (eds.), *Optimization and Design of Geodetic Networks*, Springer, 1985.

[13] F. Dacar, *An Infinite Sequence of Approximate Angle Trisections*, online manuscript, 2011, https://dis.ijs.si/france/notes/nested-trisections.pdf (accessed 5 November 2026).

[14] A. F. Porter, Trisection: The Euclidean Problem of Trisecting a Random Angle and Proof of Impossibility, *Survey Review* 31(241) (1991) 167–174, https://doi.org/10.1179/sre.1991.31.241.167.

[15] J. Wallner, R. Krasauskas and H. Pottmann, Error propagation in geometric constructions, *Computer-Aided Design* 32(11) (2000) 631–641, https://doi.org/10.1016/S0010-4485(00)00053-1.

[16] H. Desaulniers and N. F. Stewart, Robustness of numerical methods in geometric computation when problem data is uncertain, *Computer-Aided Design* 25(9) (1993) 539–545, https://doi.org/10.1016/0010-4485(93)90070-5.